

Geospatial Interpolation and Evaluation of Temperature Data over the United States

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Abstract

This study focuses on generating gridded temperature data from station-based observations using spatial interpolation techniques. Data from the United States Climate Reference Network (USCRN) were interpolated onto a 2° spatial grid and evaluated against ERA5 reanalysis data. Two interpolation methods, Inverse Distance Weighting (IDW) and Nearest Neighbor (NN), were compared using statistical metrics. The results highlight the trade-off between prediction accuracy and spatial variability representation.

1 Introduction

Geospatial interpolation is essential for transforming irregularly spaced station observations into continuous spatial fields. Such gridded datasets are widely used in climate studies and environmental modeling. In this study, temperature observations from USCRN stations are used to generate gridded datasets over the United States and evaluated against ERA5 reanalysis data.

2 Data Sources

2.1 USCRN Station Data

Station observations were obtained from the United States Climate Reference Network (USCRN), managed by NOAA. The dataset contains daily near-surface air temperature measurements from more than 130 stations.

2.2 ERA5 Reanalysis Data

ERA5 reanalysis data, developed by ECMWF, provide globally consistent climate fields. These data incorporate numerical weather prediction models and observational data assimilation, making them suitable as a reference dataset.

3 Methodology

3.1 Data Preprocessing

The station data were cleaned by removing missing values and reshaped into a long format containing station identifiers, date, temperature, and spatial coordinates.

3.2 Grid Generation

A regular grid with 2° spatial resolution was constructed over the study domain.

3.3 Interpolation Methods

3.3.1 Inverse Distance Weighting (IDW)

IDW estimates values at grid points as a weighted average of nearby observations, with weights inversely proportional to distance.

3.3.2 Nearest Neighbor (NN)

The NN method assigns the value of the closest station to each grid point without smoothing.

3.4 ERA5 Alignment

ERA5 data were processed and mapped onto the interpolation grid using nearest-neighbor search to ensure spatial alignment.

3.5 Performance Evaluation

The interpolated datasets were evaluated against ERA5 using:

- Root Mean Square Error (RMSE)
- Pearson Correlation Coefficient

3.6 Statistical Analysis

Statistical comparison included central tendency (mean), dispersion (standard deviation), and distributional similarity using the Kolmogorov-Smirnov (KS) test.

4 Results

4.1 Performance Metrics

4.2 Statistical Comparison

The IDW method produced smoother temperature fields with reduced variability, while the NN method preserved more local variability but introduced discontinuities. The KS test

Method	RMSE (K)	Correlation
IDW	9.73	0.51
Nearest Neighbor	10.77	0.60

Table 1: Performance comparison of interpolation methods

indicated moderate differences between the distributions of interpolated and ERA5 datasets.

5 Discussion

The IDW method achieved lower RMSE, indicating better performance in estimating absolute temperature values. This is due to its smoothing effect, which reduces noise in the interpolated field. However, this smoothing also leads to reduced spatial variability, reflected in a lower correlation.

In contrast, the NN method exhibited higher correlation, suggesting better preservation of spatial patterns. However, it resulted in higher RMSE and discontinuous spatial fields due to the lack of smoothing.

These results demonstrate a trade-off between accuracy and variability representation in interpolation methods. ERA5, being a physically-based reanalysis dataset, provides more consistent spatial fields compared to purely statistical interpolation methods.

6 Model Suitability

Based on the evaluation, IDW is more suitable for generating gridded temperature datasets due to its lower prediction error and smoother spatial representation. NN may be useful as a baseline method but is less reliable for practical applications.

7 Limitations

- Uneven spatial distribution of stations
- Lack of physical constraints in interpolation methods
- Differences between observational data and reanalysis datasets

8 Conclusion

This study demonstrates the application of spatial interpolation techniques to generate gridded temperature datasets. IDW provides a reasonable approximation of spatial temperature fields, while NN highlights the importance of smoothing in interpolation. Future work may include advanced geostatistical methods such as kriging to improve accuracy.